Walmart Sales Dashboard: A Data-Driven Study of Retail Sales, Store Performance, and Economic Factors

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Problem Statement

Retail organisations generate large volumes of transactional data across stores, time periods, and economic conditions. Analysing patterns in weekly sales, the impact of holidays, the effect of fuel prices, and temperature on consumer behaviour is challenging when data remains in its raw CSV form. This project aims to answer key questions such as:

- How do weekly sales vary across Walmart's 45 stores over 2010–2012?
- Which stores are the top and bottom performers by total revenue?
- What percentage of sales occur during holiday vs non-holiday weeks?
- Is there a relationship between fuel price and weekly sales?
- How does average temperature vary across store locations?
- What is the overall financial scale of Walmart's retail operations?

The objective is to transform raw retail data from the Kaggle Walmart Sales Dataset into meaningful, interactive insights using Power BI visualisations.

Features

This project presents a Power BI report consisting of a single-page interactive dashboard with multiple visual components, each connected to Year and Store slicers for dynamic filtering:

- Key KPI Cards:
  - Total Sales: 6.74bn
  - Avg Weekly Sales: 1.05M
  - Total Stores: 45
  - Holiday Sales: 505.30M
- Total Sales by Month – Line Chart showing monthly sales trend; sales peak in July and dip in December.
- Total Sales by Store – Clustered Bar Chart across all 45 stores; Store 20 is the top performer at 301M.
- Holiday vs Non-Holiday – Donut Chart showing 92.5% of sales occur during non-holiday weeks.
- Top Stores Table – Store, Total Sales, and Avg Weekly Sales; Store 20 leads with \$2.1M average weekly sales.

- Fuel Price vs Sales – Scatter Chart across all stores; reveals weak negative correlation between fuel price and total sales.
- Avg Temperature by Store – Column Chart showing temperature varies from 40°F to 70°F across store locations.
- Year Slicer – checkbox style (2010, 2011, 2012) updating all visuals simultaneously.
- Store Slicer – dropdown (default: All) for filtering to individual store performance.

Data Transformation

Raw data from the Kaggle Walmart Sales Dataset required the following cleaning and transformation steps in Power Query before visualisation:

- Date column type corrected using English (United Kingdom) locale to resolve DD-MM-YYYY parsing errors.
- All numeric columns (Weekly_Sales, Temperature, Fuel_Price, CPI, Unemployment) set to Decimal Number type.
- Holiday_Flag and Store columns set to Whole Number type.
- Month Name column added via Add Column → Date → Month → Name of Month.
- Month Number column added for correct calendar sorting of Month Name.
- Month Name sorted by Month Number column using Sort by Column in Modeling tab.
- Holiday Type calculated column created using DAX IF() to label 0/1 as Non-Holiday / Holiday.
- Store column changed to Text type in Modeling tab to prevent incorrect aggregation.

DAX Measures Created

- **Total Sales:** SUM('Walmart'[Weekly_Sales])
- **Avg Weekly Sales:** AVERAGE('Walmart'[Weekly_Sales])
- **Total Stores:** DISTINCTCOUNT('Walmart'[Store])
- **Holiday Sales:**
CALCULATE(SUM('Walmart'[Weekly_Sales]),
'Walmart'[Holiday_Flag] = 1)
- **Non Holiday Sales:**
CALCULATE(SUM('Walmart'[Weekly_Sales]),
'Walmart'[Holiday_Flag] = 0)

Tech Stack

- Power BI Desktop – Data transformation, DAX measure creation, and interactive report design
- Kaggle Walmart Dataset – Data source: 1,000 rows, 17 columns, 45 stores, 2010–2012
- DAX (Data Analysis Expressions) – SUM, AVERAGE, CALCULATE, DISTINCTCOUNT, DIVIDE, IF
- Power Query (M Language) – Date type correction, locale fix, column additions

- Microsoft Excel – Initial data inspection before loading into Power BI
- GitHub – Public repository for project version control and submission

Unique Points

This dashboard demonstrates several advanced Power BI techniques and provides multi-dimensional retail insights:

Multi-dimensional insights:

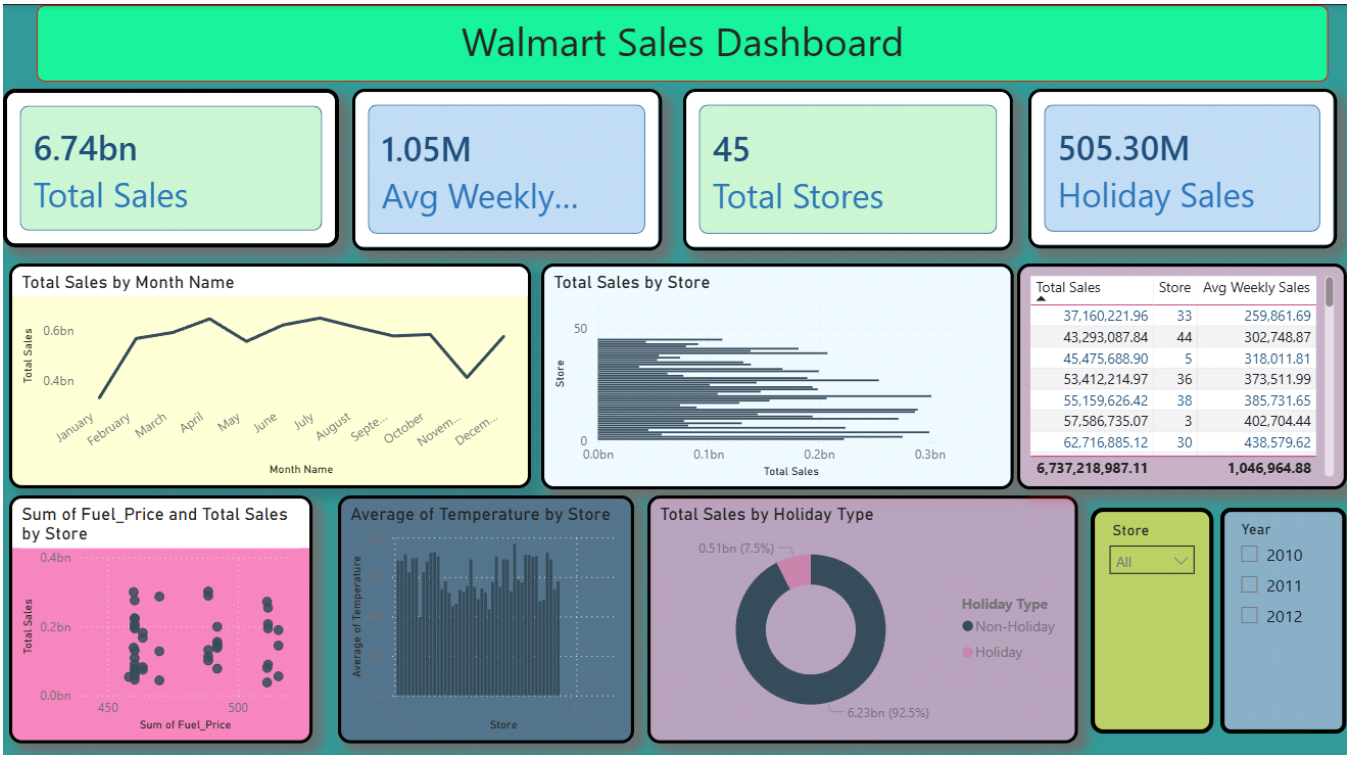
- Comprehensive analysis covering 3 years of US retail data (2010-2012) across 45 Walmart stores and 1,000 weekly transaction records.
- Economic factor analysis – Fuel Price vs Sales scatter chart surfaces a relationship between external economic conditions and retail performance.
- Geographic insight – Average Temperature by Store column chart reveals the physical diversity of store locations across the US, ranging from 40°F to 70°F.
- Holiday segmentation – DAX-calculated Holiday Type column cleanly separates holiday and non-holiday weeks, confirming 92.5% of sales are driven by consistent non-holiday demand.

Use of advanced techniques:

- Locale-specific date parsing fix – real-world data pipeline challenge resolved using English (United Kingdom) locale in Power Query to correctly parse DD-MM-YYYY dates.
- Month Name sort fix using Sort by Column – resolves the common Power BI issue of months appearing in alphabetical rather than calendar order.
- Six distinct visual types in one layout: KPI Cards, Line Chart, Clustered Bar Chart, Scatter Chart, Column Chart, Donut Chart, and Table – demonstrating broad coverage of Power BI's visualisation library.
- Dual slicers (Year + Store) allow stakeholders to analyse any combination of year and individual store without technical expertise.
- Store column intentionally converted to Text type in Modeling tab to prevent Power BI from incorrectly treating store numbers as numerical aggregates.

Future Improvements

- Integration of all 45-store weekly data from the full Kaggle Walmart Forecasting Competition dataset (currently using the 1,000-row sample).
- Addition of a US Map visual plotting Total Sales by Store location to geographically represent store performance across the country.
- Inclusion of a CPI vs Sales scatter chart to analyse how inflation (Consumer Price Index) correlates with weekly revenue.
- Unemployment Rate trend line overlaid on the Sales by Month chart to surface macroeconomic impact on consumer spending patterns.

• Conditional formatting on the Top Stores table to highlight the top 3 performers in green and bottom 3 in red for instant identification.
• Drill-through detail pages per store reachable by right-clicking any bar – showing that store's full monthly sales history and KPIs.
• Addition of a Sales Forecast visual using Power BI's built-in Analytics pane to project future weekly sales trends based on historical data.
• Optimisation of the dashboard for Power BI's Phone Layout view to maintain full readability on mobile devices for on-the-go stakeholder access.

Screenshot